APRIL 2025

53601/126C2A/
127C2A/225C3A/
220C4A/241C4A

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer any TEN questions.

1. What is Object-Oriented Programming?

2. What is the purpose of the main method in a Java program?

3. Identify the primary difference between String and String Buffer classes in Java.

4. Define a user-defined class in Java.

5. Identify the different types of inheritance in Java.

6. What is the purpose of the finally block in exception handling?

7. Identify the difference between throw and throws in Java.

8. What is the purpose of the AWT class hierarchy in Java?

9. Define a top-level container in Swing.

10. What is the function of the JLabel component in Swing?

11. What is an adapter class in Java?

12. What is the role of the Comparator interface?

PART B — ($5 \times 5 = 25$ marks)

Answer any FIVE questions.

13. Describe the architecture of the Java Virtual Machine (JVM) and its significance in executing Java programs.

14. Explain the different data types available in Java.

15. Explain the concept of method overloading in Java with an example.

16. Examine the concept of packages in Java.

17. Explain the process of creating your own exception classes in Java.

18. Explain the hierarchy of Swing components with an example.

19. What are inner classes in Java? Explain their different types.

PART C — ($3 \times 10 = 30$ marks)

Answer any THREE questions.

20. Evaluate the role of static data and static methods in Java. Discuss their benefits and limitations.

21. Analyze the different types of inheritance in Java, including their benefits and drawbacks.

22. Discuss the concept of deadlock in multithreaded programming.

23. Explain the role of containers in Swing.

24. Discuss the use and advantages of adapter classes in Java's event-handling mechanism.

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